EG1311: Design & Make

Design & Prototyping

Dr. Jason S. Ku

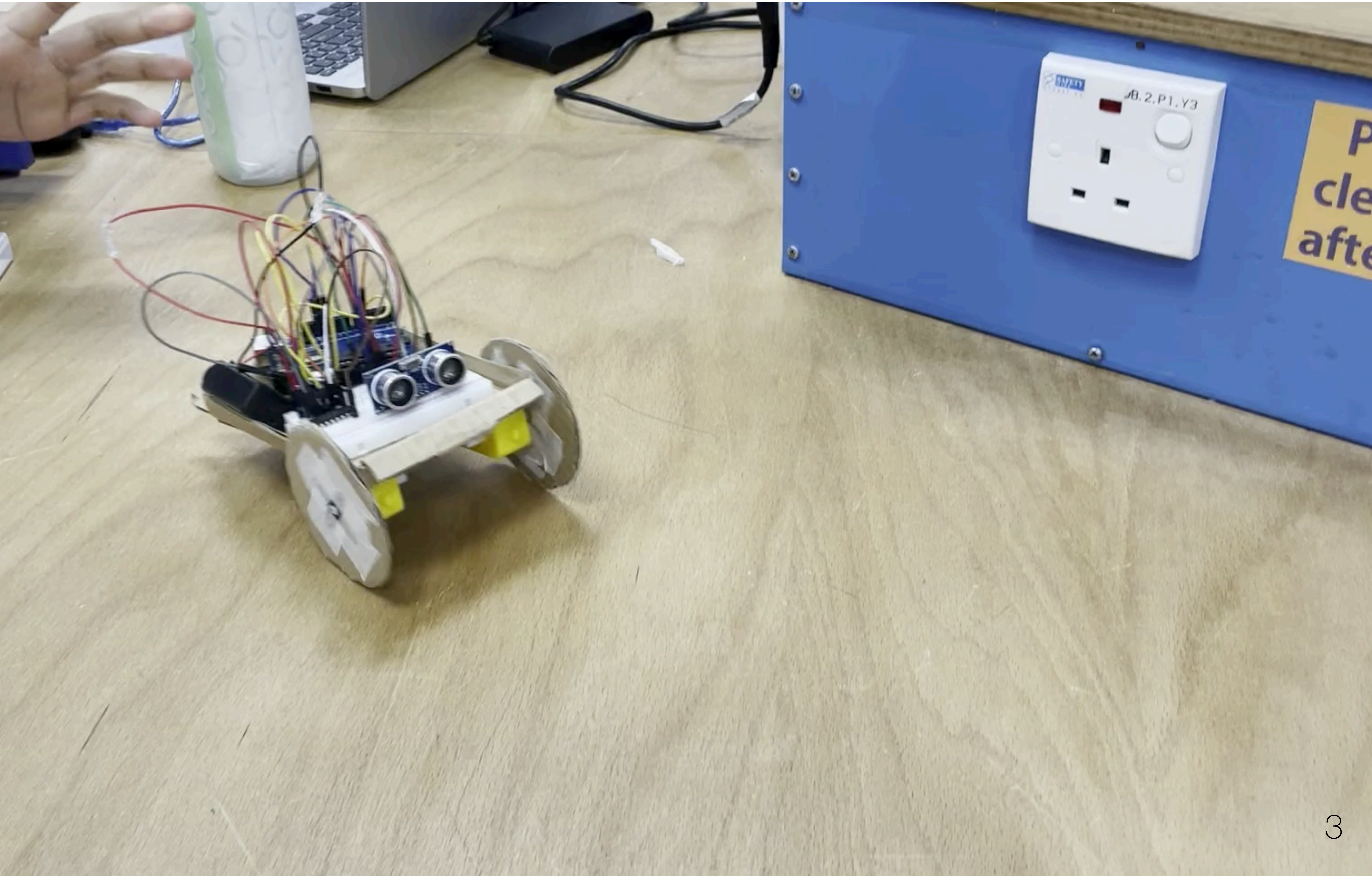
NUS National University of Singapore | College of Design and Engineering

jasonku@nus.edu.sg

Skills

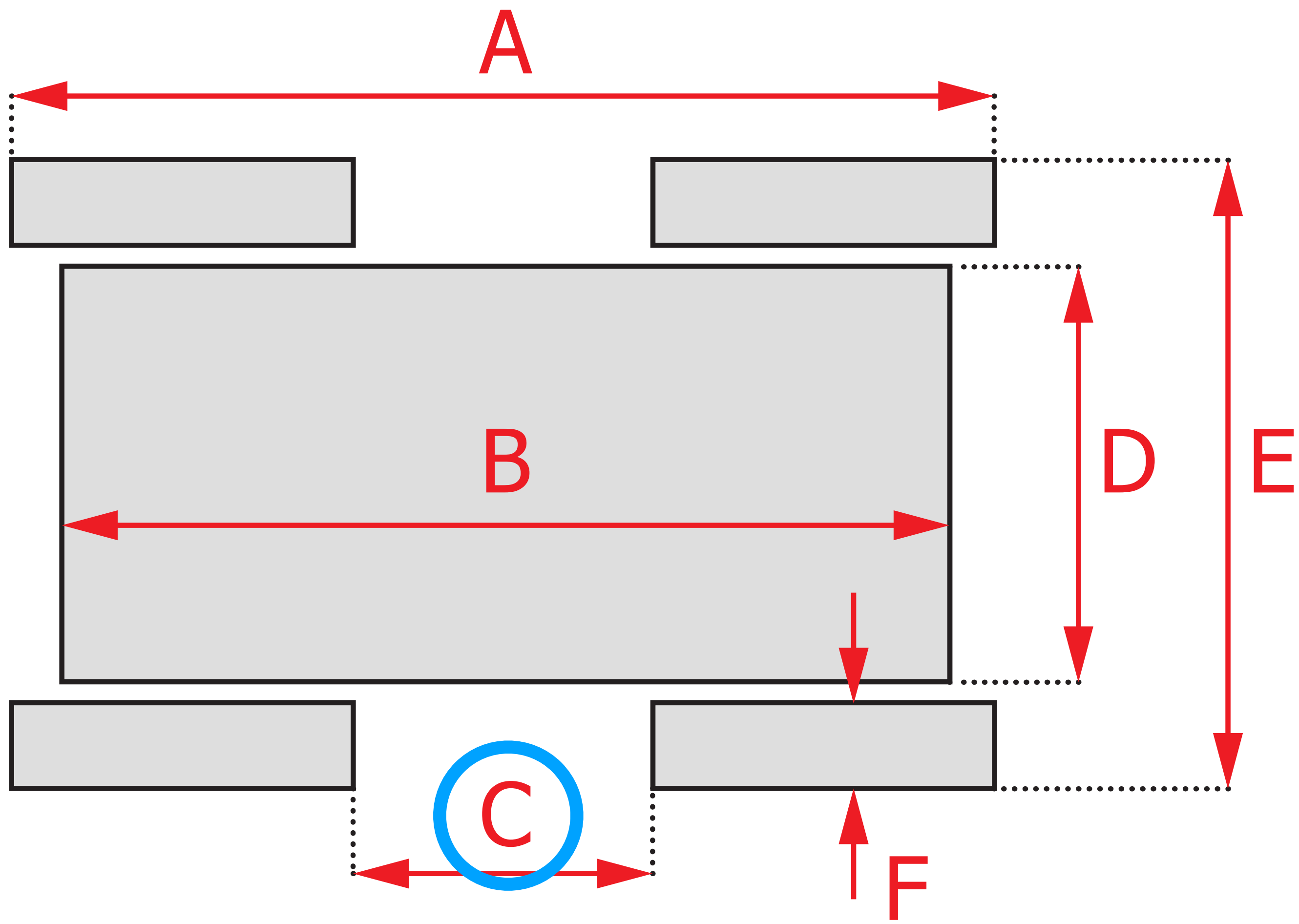
- Electronics
  - Computers / Programming
  - Arduino Hardware / Software
  - Sensor / Serial
  - Motor / Driver / Servo
- CAD
  - Sketches / Constraints
  - Part Modelling / Assemblies
  - Animation / Appearance / Drawing
  - Measurement

Template Project



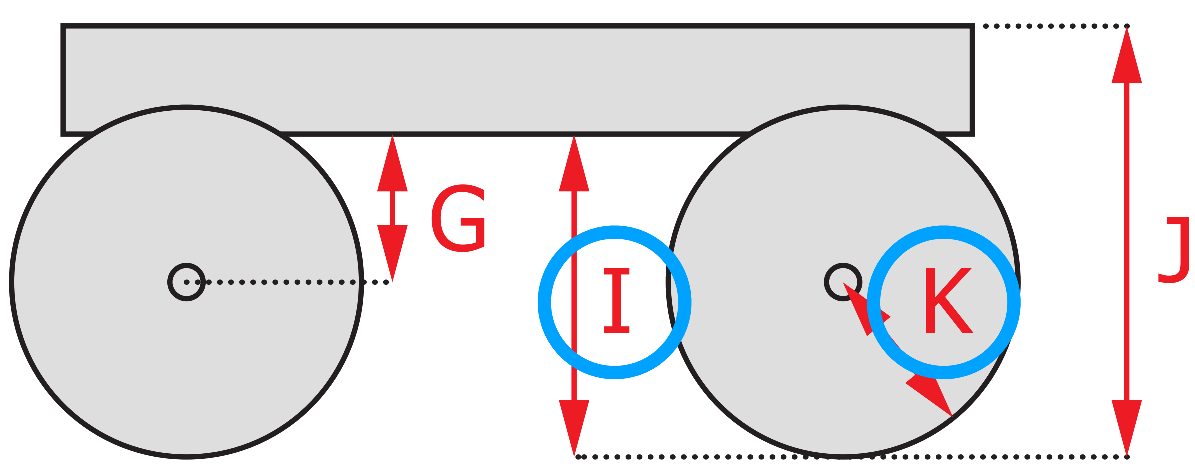
Design Parameters

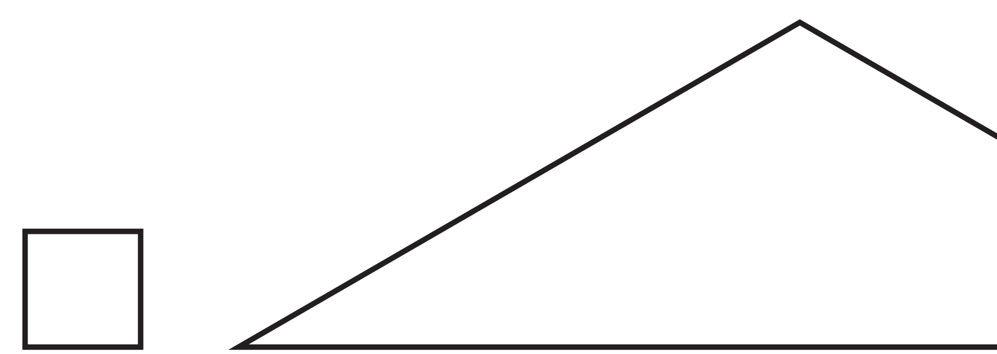
Top



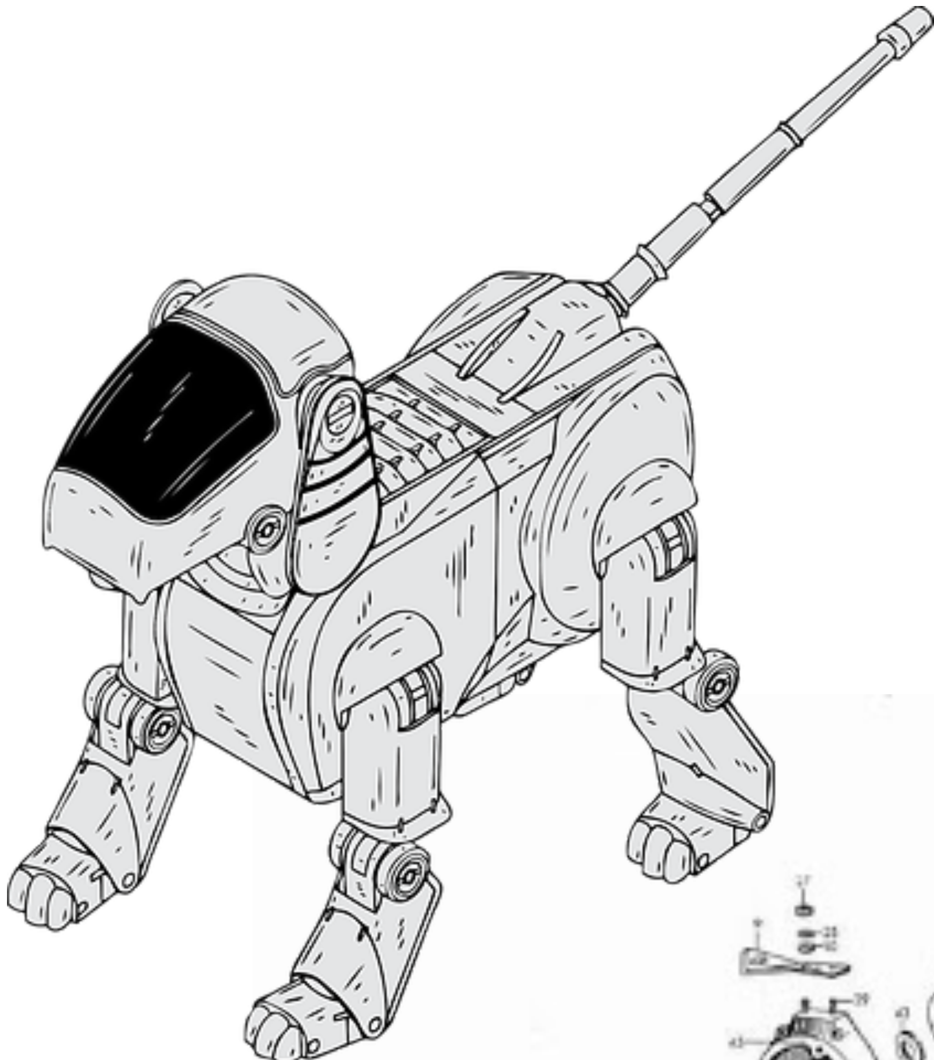
- Wheels
- Power
- Friction

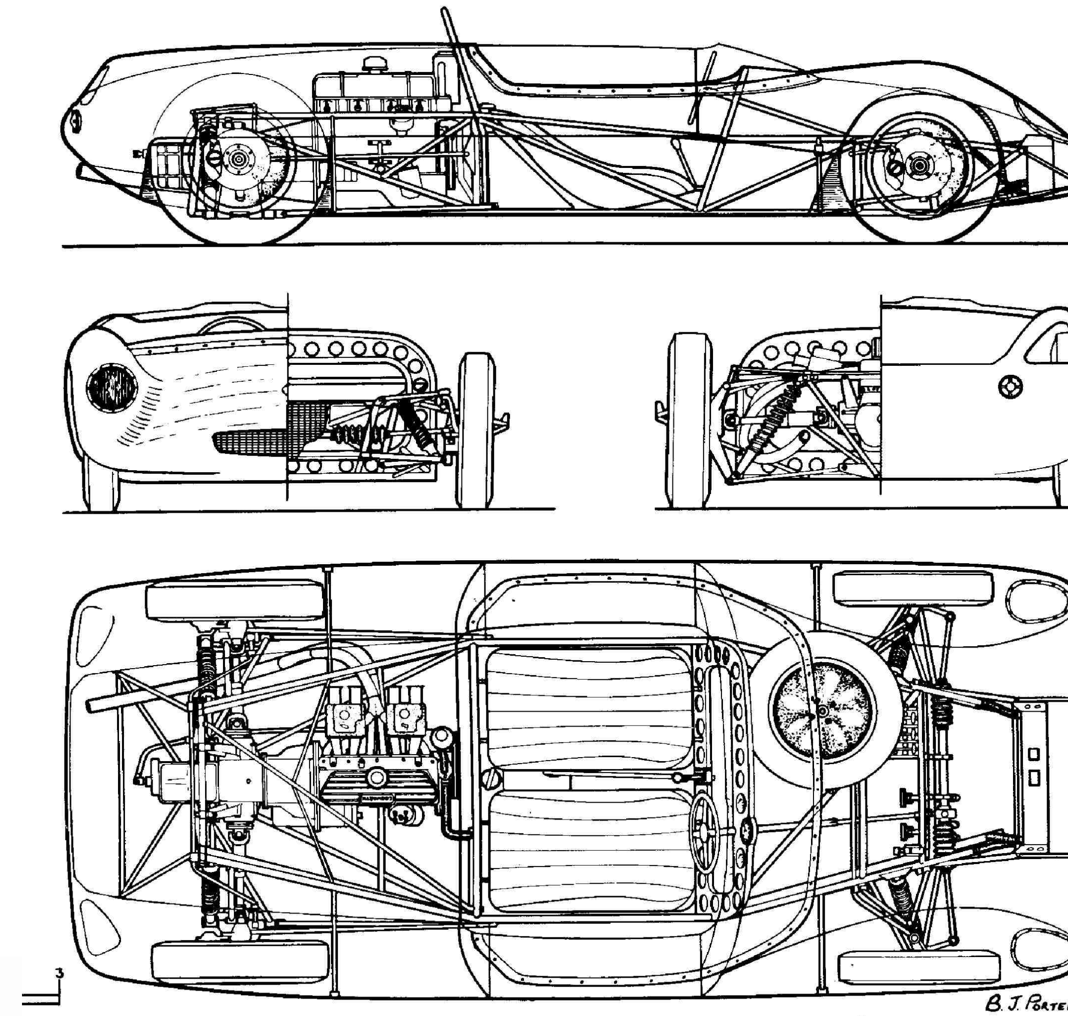
Side

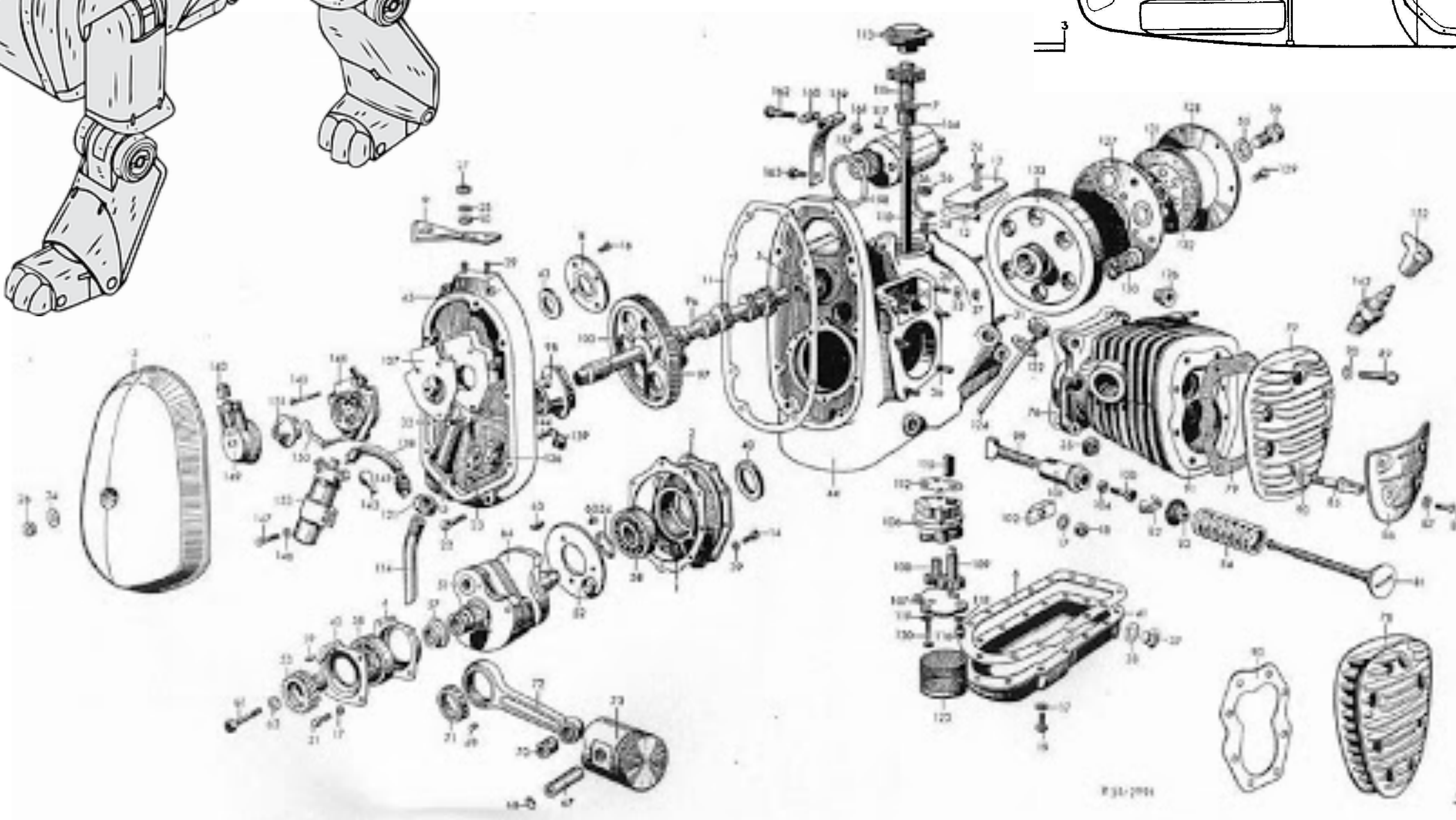




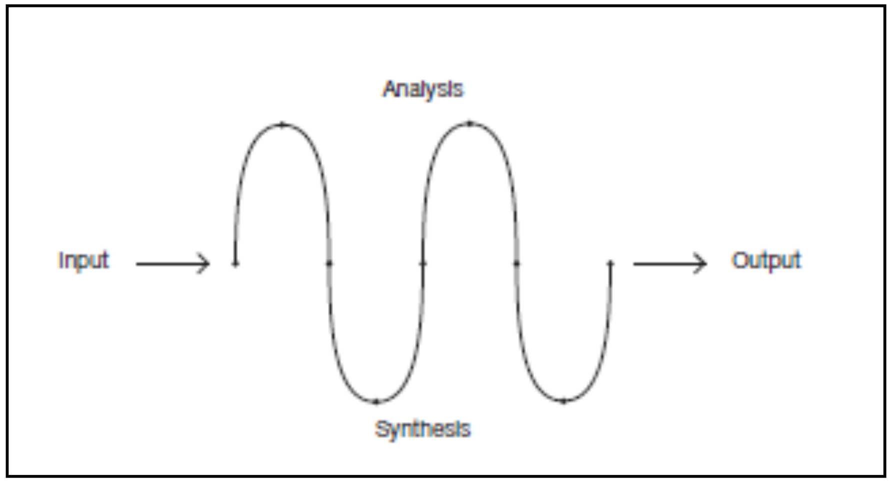
How to design?

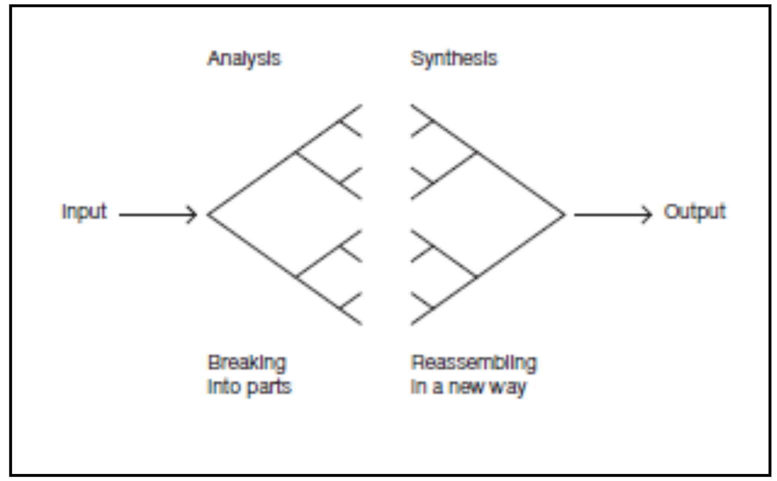


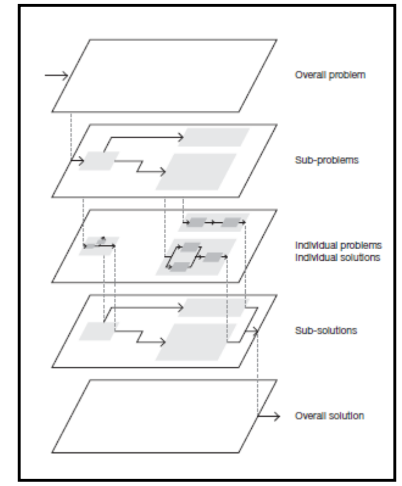


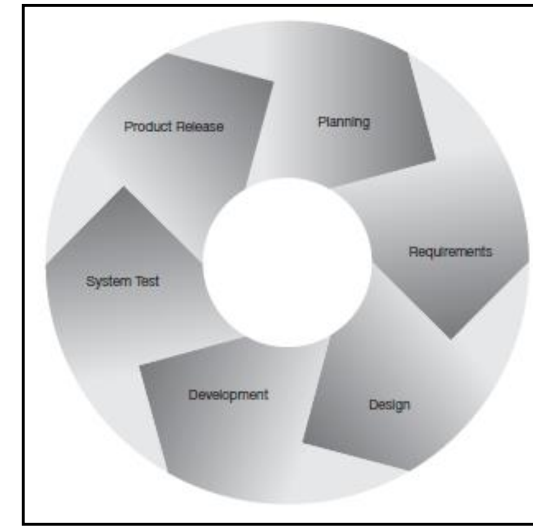


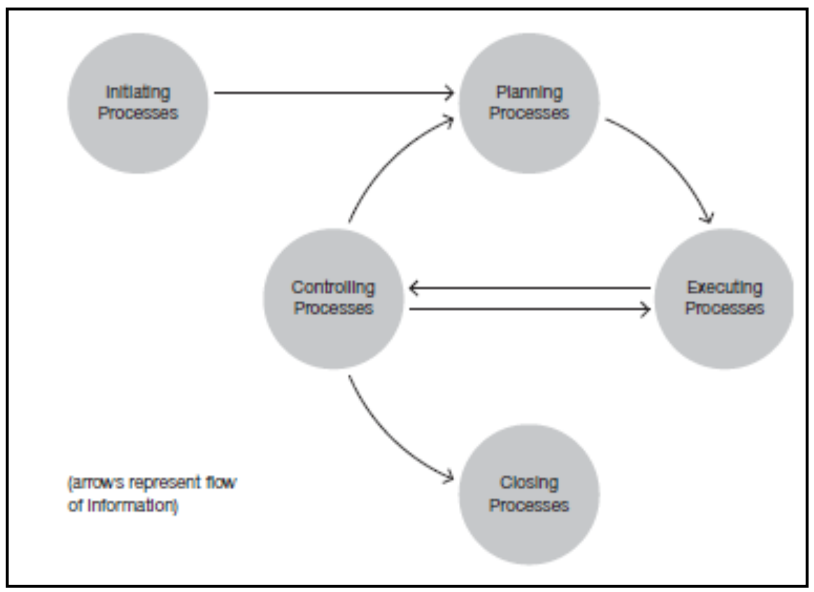
Design Processes

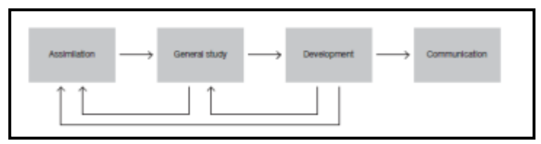


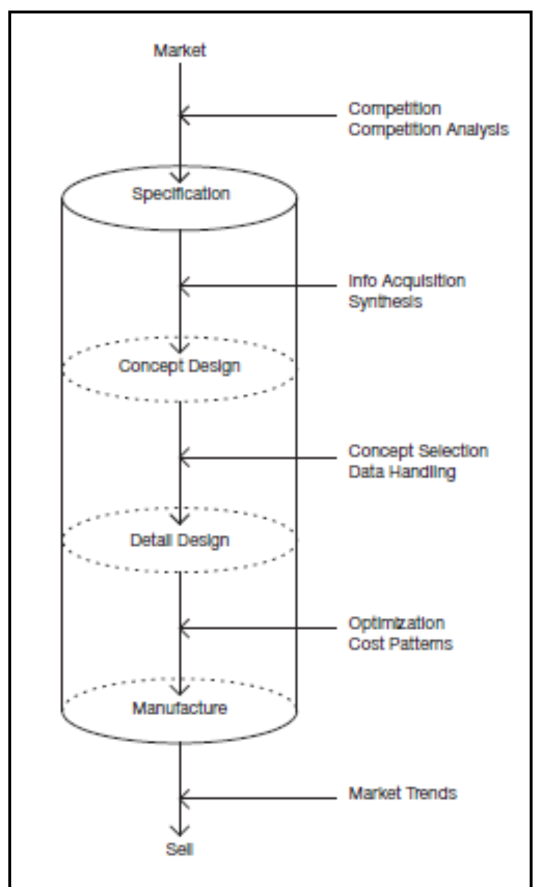


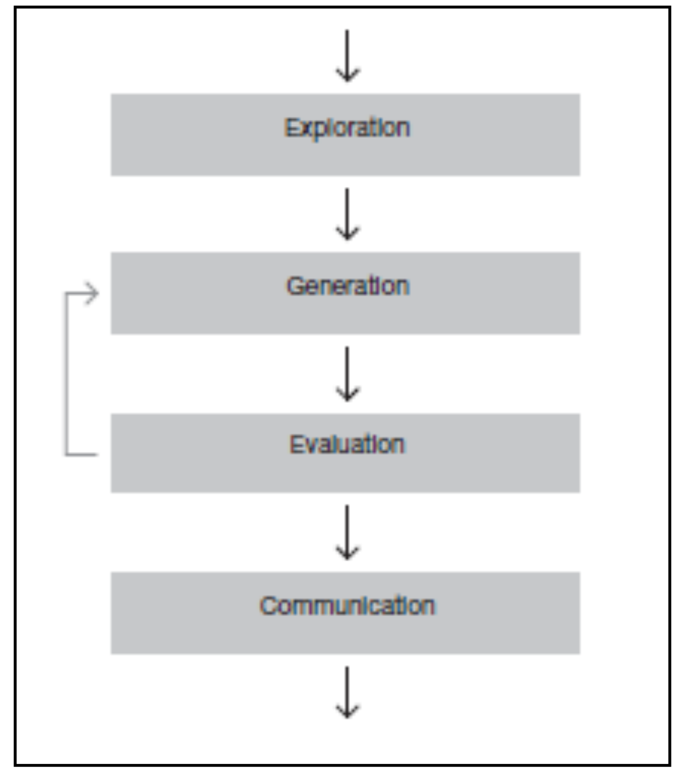


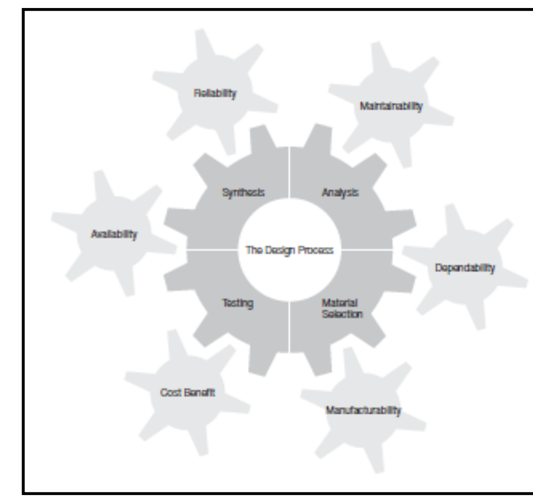


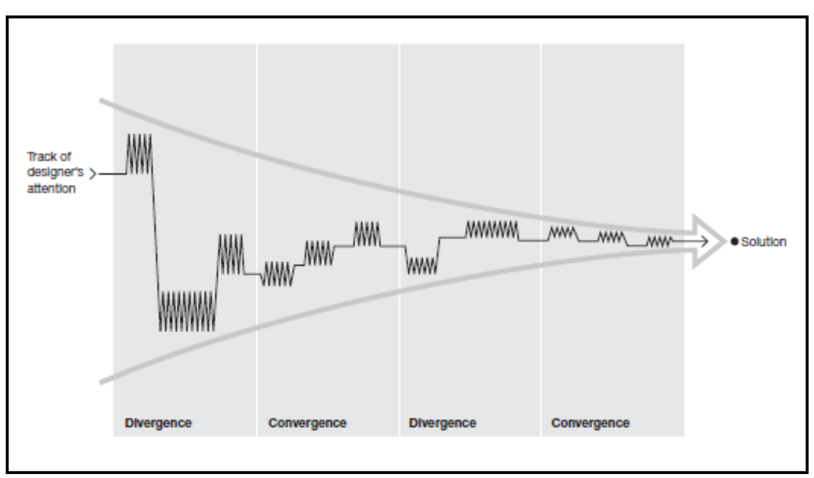


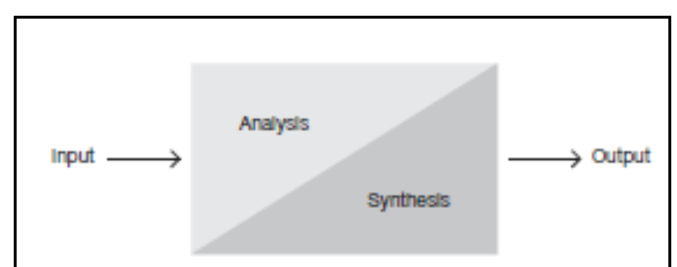


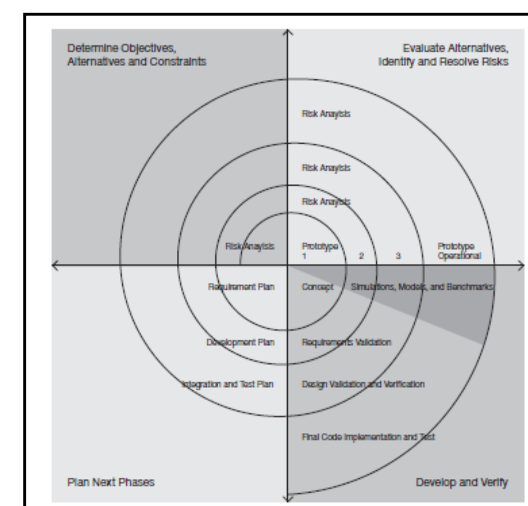






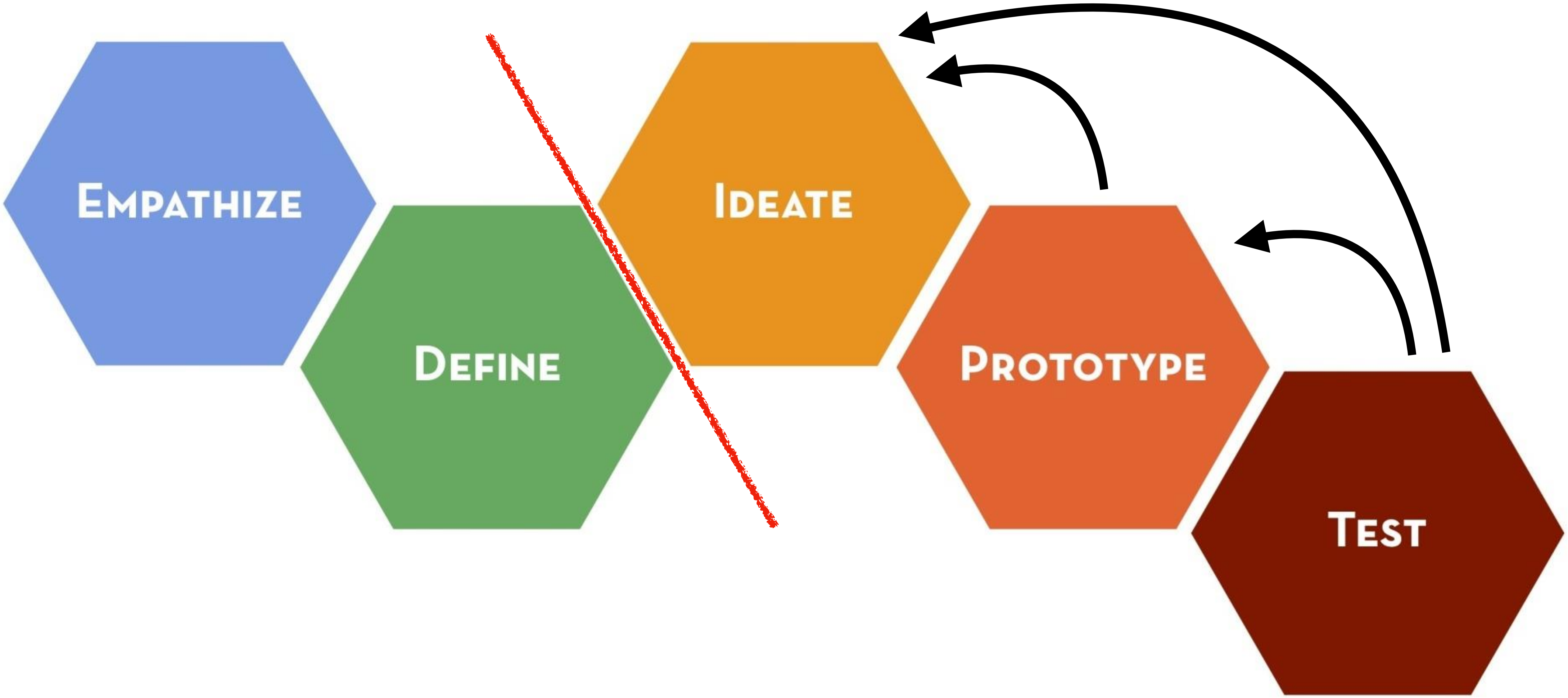






Adapted: *How do you design?* by Hugh Dubberly (Dubberly Design Office)

Design Thinking



Adapted: *Bootcamp Bootleg* (Institute of Design at Stanford)

Brainstorming

- Ask provocative questions (what if…)
- Go for **quantity** (“quantity breeds quality”)
- Build on the ideas of others
- Criticism postponed

Ideation Loop

1. Set a topic
2. Individually generate many ideas
3. Present those ideas to each other
4. Give positive feedback, criticism postponed
5. Repeat steps 2-4 as needed

Communication!

Iteration & Testing

https://www.youtube.com/watch?v=H0_yKBitO8M

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